Texas Instruments Placement Paper May 2011:-

1. Minimum number of 2-input NAND gates that will be required to implement the function: Y = AB + CD + EF is
a. 4
b. 5
c. 6
d. 7

2. Consider a two-level memory hierarchy system M1 & M2. M1 is accessed first and on miss M2 is accessed. The access of M1 is 2 nanoseconds and the miss penalty (the time to get the data from M2 in case of a miss) is 100 nanoseconds. The probability that a valid data is found in M1 is 0.97. The average memory access time is:
a. 4.94 nanoseconds
b. 3.06 nanoseconds
c. 5.00 nanoseconds
d. 5.06 nanoseconds

3. Interrupt latency is the time elapsed between:
a. Occurrence of an interrupt and its detection by the CPU
b. Assertion of an interrupt and the start of the associated ISR
c. Assertion of an interrupt and the completion of the associated ISR
d. Start and completion of associated ISR

4. Which of the following is true for the function (A.B + A.C + B.C)
a. This function can glitch and can be further reduced
b. This function can neither glitch nor can be further reduced
c. This function can glitch and cannot be further reduced
d. This function cannot glitch but can be further reduced

5. For the two flip-flop configuration below, what is the relationship of the output at B to the clock frequency?
a. Output frequency is 1/4th the clock frequency, with 50% duty cycle
b. Output frequency is 1/3rd the clock frequency, with 50% duty cycle
c. Output frequency is 1/4th the clock frequency, with 25% duty cycle
d. Output frequency is equal to the clock frequency

XOR

A B

Q
D Q

D

CLK

CLK
Q

Q

6. The voltage on Node B is:
a. 0
b. 10
c. 10
d. 5

10O
10O 10O

+ +

10O
10O
10V 20V
– –

GND B

7. A CPU supports 250 instructions. Each instruction op-code has these fields: The instruction type (one among 250) A conditional register specification 3 register operands Addressing mode specification for both source operands The CPU has 16 registers and supports 5 addressing modes. What is the instruction op-code
length in bits?
a. 32
b. 24
c. 30
d. 36

8. In the iterative network shown, the output Yn of any stage N is 1 if the total number of 1s at the inputs starting from the first stage to the Nth stage is odd. (Each identical box in the iterative network has two inputs and two outputs). The optimal logic structure for the box consists of:
a. One AND gate and one NOR gate

b. One NOR gate and one NAND gate
c. Two XNOR gates
d. One XOR gate

I 1 I 2 I n I n +1 I n + 2
0
Y1 Y2 Yn Yn+1 Yn+2

9. Consider a circuit with N logic nets. If each net can be stuck-at either values 0 and 1, in how many ways can the circuit be faulty such that only one net in it can be faulty, and such that up-to all nets in it can be faulty?
a. 2 and 2N
b. N and 2^N
c. 2N and 3^N-1
d. 2N and 3N

10. if a 5-stage pipe-line is flushed and then we have to execute 5 and 12 instructions respectively then no. of cycles will be
a. 5 and 12
b. 6 and 13
c. 9 and 16
d.none

11. k-map

ab
---
c 1 x 0 0
1 x 0 x

solve it

a. A.B
B. ~A
C. ~B
D. A+B

12.Char A[10][15] and INT B[10][15] is defined what is the address of A[3][4] abd B[3][4] if address of OD A is OX1000 and B is 0X2000

A. 0X1030 AND 0X20C3
B. OX1031 AND OX20C4

13. int f(int *a)

```
{
int b=5;
a=&b;
}
main()
{
int i;
printf("\n %d",i);
f(&i);
printf("\n %d",i);
}
```

what's the output .

1.10,5
2,10,10
c.5,5
d. none

14.
```
main()
{
int i;
fork();
fork();
fork();
printf("----");
}
```

how many times the printf will be executed .
a.3
b. 6
c.5
d. 8

15.
```
void f(int i)
{
int j;
for (j=0;j<16;j++)
{
if (i & (0x8000>>j))
printf("1");
else
printf("0");
}
```

```
}
```
what's the purpose of the program

a. its output is hex representation of i
b. bcd
c. binary
d. decimal

16.
```
#define f(a,b) a+b
#define g(a,b) a*b

main()
{

int m;
m=2*f(3,g(4,5));
printf("\n m is %d",m);
}
```

what's the value of m
a.70
b.50
c.26
d. 69

17.
```
main()
{
char a[10];
strcpy(a,"\0");
if (a==NULL)
printf("\a is null");
else
printf("\n a is not null");}
```

what happens with it .
a. compile time error.
b. run-time error.
c. a is null
d. a is not null.